Predictive Analytics for Proactive Order Fulfilment Risk Management

Table of Contents

1.Introduction 1
2. Problem Definition 2
3. Aim and Objectives 2
4. Scope 3
5. Methodology 3
6. Data Sources 5
7. Technologies and Tools 5
8. Expected Deliverables 6
9. Project Timeline 6
10. Ethical and Legal Considerations 7
11. Conclusion 7
References 7

1.Introduction

Order fulfilment is an important part of supply-chain operations. When orders are delayed, it can lead to unhappy customers and extra work for the teams responsible for managing deliveries. If there was a way to identify orders that might be delayed before the problem happens, fulfilment teams could have more time to investigate and decide whether any action is needed.

I chose this project because it relates to my current role as an IT Product Manager supporting Supply Chain Fulfilment. Working with both business and technology teams has given me an interest in how data can be used to improve processes and help people make better decisions. Recent research suggests that predictive analytics and modern technologies such as Internet of Things (IoT) systems and data analytics tools can play a significant role in improving supply-chain performance and supporting data-driven decision-making for continuous improvement (Sy, 2025). I also see this project as an

opportunity to put what I have learned during the Higher Diploma in Data Analytics for Business into practice.

The project will use historical supply-chain data to investigate whether machine learning can help predict which orders are at risk of late delivery. I want to find out whether the data contains useful patterns that could help identify potential delays earlier, rather than relying only on manual checks or waiting until an order is already late.

2. Problem Definition

Late deliveries can happen for different reasons, such as the type of product ordered, the shipping method and where the order is being delivered. I want to find out if looking at this information can help identify which orders might arrive late, giving fulfilment teams more time to look into potential problems.

By comparing different machine learning models, I hope to find out whether historical order and shipping information can help identify orders that may be delayed. I will then look at how the results could be useful to fulfilment teams when deciding which orders may need further attention.

3. Aim and Objectives

Aim

The aim of this project is to develop and evaluate machine learning classification models to predict late-delivery risk using historical supply-chain data.

The project will examine which variables are useful for making predictions and compare different models to see which performs best.

Objectives

The main objectives are:

1. Collect, clean and prepare the dataset for analysis.
2. Explore the data to identify patterns relating to late deliveries.
3. Select and create suitable variables for predicting late-delivery risk.
4. Develop and compare different machine learning classification models.
5. Evaluate the models using appropriate performance measures.

Business Objective

The business objective is to investigate whether predictive analytics could help fulfilment teams identify potentially delayed orders earlier and prioritise which orders may need further investigation.

4. Scope

This project will use historical supply-chain data to predict late-delivery risk based on order, product, geographical and shipping characteristics. It will include data preparation, exploratory analysis, feature engineering, development and evaluation of Logistic Regression, Decision Tree and Random Forest models, and recommendations based on the findings.

The project will focus on predicting whether an order is at risk of late delivery. It will exclude exact delivery-time prediction, investigation of individual delay causes, warehouse and inventory optimisation, delivery route and carrier allocation optimisation, and financial forecasting. The analysis will use historical data rather than real-time order tracking.

5. Methodology

Phase 1: Data Collection

Historical supply-chain data will be obtained from the DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS dataset, available at https://doi.org/10.17632/8gx2fvg2k6.5.

Phase 2: Data Preparation

Python will be used to clean and prepare the dataset. Tasks will include:

- Renaming columns where needed
- Removing unnecessary variables
- Handling missing values and duplicates
- Correcting data types and inconsistencies
- Removing unnecessary personal information

Phase 3: Exploratory Data Analysis

Exploratory data analysis will be used to identify patterns and relationships associated with late deliveries. This will include:

- Late-delivery risk distribution
- Shipping mode analysis
- Product category and market comparisons
- Customer segment and order quantity analysis

Charts and visualisations will help identify useful patterns and potential predictors.

Phase 4: Feature Engineering

Relevant variables will be selected and transformed to improve model performance. New features, such as order month or day of the week, may also be created to test if it will improve performance.

Phase 5: Machine Learning Models

The following classification algorithms will be developed and compared:

- Logistic Regression
- Decision Tree
- Random Forest

Logistic Regression will be used as the baseline classification model. Decision Tree will be used because it can identify non-linear relationships and interactions between variables. Random Forest will be used because combined learning methods often provide improved predictive performance and reduced overfitting compared to individual decision trees.

The dataset will be divided into training and testing sets to evaluate model performance on unseen data.

Phase 6: Model Evaluation

The models will be evaluated and compared using the following metrics:

- Accuracy.
- Precision.
- Recall.
- F1-score.
- Confusion matrix.

Particular attention will be given to recall, as identifying orders at risk of late delivery is a key objective.

Phase 7: Visualisation and Recommendations

The final stage will present the results through charts, model comparisons and an evaluation of important variables. Recommendations will be made on how the findings could support fulfilment planning and help identify potentially late deliveries.

6. Data Sources

The dataset selected for this project is *DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS*, obtained from Mendeley Data (Constante, Silva and Pereira, 2019). It contains approximately 180,500 order records and 53 variables covering supply-chain information such as order details, product categories, customer segments, shipping methods and delivery outcomes.

Dataset link: https://data.mendeley.com/datasets/8gx2fvg2k6/5

The dataset is suitable for this project because it provides information that can be analysed to investigate patterns associated with late deliveries. The Late_delivery_risk variable will be used as the target for the classification models, while relevant order, product, geographical and shipping variables will be considered as potential predictors. The data will be cleaned and prepared before analysis, with unnecessary identifying information removed and variables that could cause data leakage excluded.

7. Technologies and Tools

The following tools will be used throughout the project:

- Python: Main programming language.
- Pandas and NumPy: Data cleaning and preparation.
- Matplotlib and Seaborn: Data visualisation and exploratory analysis.
- Scikit-learn: Machine learning model development and evaluation.
- Jupyter Notebook: Analysis and documentation.

8. Expected Deliverables

By the end of the project, I expect to have completed the following:

- Data Preparation

  A cleaned dataset with unnecessary columns removed and any data-quality issues documented.

- Exploratory Data Analysis

  An analysis of the dataset showing patterns and relationships that may help explain late-delivery risk.

- Predictive Features

  A set of selected variables suitable for training the classification models, with attention given to avoiding data leakage.

- Machine Learning Models

  Three classification models developed using Logistic Regression, Decision Tree and Random Forest.

- Model Evaluation

  A comparison of the models using accuracy, precision, recall, F1-score and confusion matrices.

- Final Report

A report explaining the business problem, the dataset, the methodology, the modelling process, the results and the main conclusions.

- Visual Presentation

  Charts or a Power BI dashboard, where appropriate, to present the findings in a clear and understandable way.

  The final outcome should provide a clearer understanding of whether historical supply-chain data can help identify orders that may be at risk of late delivery.

9. Project Timeline

The project will be completed over the course of the second semester. The work will be organised around the main stages of the analysis, starting with data collection and preparation, followed by exploratory analysis, model development and evaluation.

The final stage will focus on interpreting the results, preparing any visualisations and completing the project report and presentation.

The timeline may need to be adjusted as the project progresses, especially if the dataset requires more preparation than expected or if changes are needed to the modelling approach.

Milestone	**Target Month**
Proposal	October
Data Prepared	December
Features Finalised	January
Models Complete	March
Evaluation Complete	April
Submission	April/ May

10. Ethical and Legal Considerations

The DataCo dataset is available under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence. I will acknowledge the dataset source and authors appropriately. As the dataset contains customer-related information, unnecessary identifiers, such as names, email addresses and street addresses will be removed, and only relevant variables will be retained. The data will be stored securely, and identifying information will not be included in the final report or presentation. I will consider relevant GDPR and Irish data-protection requirements and follow CCT College Dublin's ethics procedures, including confirming whether additional approval is required.